

Free energy and current

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1 Free energy

1.1 Grand canonical potential

In the grand canonical ensemble, the grand partition function is:

$$\mathcal{Z} = \sum_{\{n_i\}} e^{-\beta(E - \mu N)}$$

Here:

- $\{n_i\}$ is a configuration of occupation numbers of single-particle states,
- $E = \sum_i n_i \varepsilon_i$, total energy of a configuration,
- $N = \sum_i n_i$, total number of particles,
- μ is the chemical potential,
- Fermions: $n_i = 0$ or 1 .

1.1.1 Why non-interacting matters

Since the fermions are non-interacting, the total energy is simply the sum of the individual occupied energy levels:

$$E - \mu N = \sum_i n_i (\varepsilon_i - \mu)$$

So the Boltzmann weight becomes:

$$e^{-\beta(E - \mu N)} = \prod_i e^{-\beta n_i (\varepsilon_i - \mu)}$$

This factorizes across all i — this is crucial.

1.1.2 Sum over occupations

Now we sum over all configurations of $\{n_i\}$, with $n_i = 0$ or 1:

$$\mathcal{Z} = \sum_{n_1=0}^1 \sum_{n_2=0}^1 \dots \prod_i e^{-\beta n_i (\varepsilon_i - \mu)} = \prod_i \sum_{n_i=0}^1 e^{-\beta n_i (\varepsilon_i - \mu)}$$

Each sum is just over two terms:

$$\sum_{n_i=0}^1 e^{-\beta n_i (\varepsilon_i - \mu)} = 1 + e^{-\beta (\varepsilon_i - \mu)}$$

So the total partition function becomes:

$$\boxed{\mathcal{Z} = \prod_i \left(1 + e^{-\beta (\varepsilon_i - \mu)} \right)}$$

This product form comes directly from:

- The non-interacting assumption, and
- The fact that fermionic occupation numbers are 0 or 1.

To derive the expression for the **free energy per edge**:

1.2 General expression for free energy

For a system of non-interacting fermions, the grand potential Ω at temperature T and chemical potential μ is given by:

$$\Omega = -k_B T \sum_i \ln \left[1 + e^{-(\varepsilon_i - \mu)/k_B T} \right],$$

and the Helmholtz free energy F is related to Ω via:

$$F = \Omega + \mu N.$$

But if $\mu = 0$ (often the case at particle-hole symmetry or neutrality), then $F = \Omega$. Also, in our case, the energy spectrum is symmetric around 0, and $\varepsilon_i > 0$, and the integral is over positive energies.

1.2.1 Connection to density of states

The sum over energy levels becomes an integral using the density of states $\rho^\sigma(\epsilon, \phi^\sigma)$:

$$F_{\text{bulk}}^\sigma = -k_B T \int_0^\infty d\epsilon \rho^\sigma(\epsilon, \phi^\sigma) \ln \left[1 + e^{-\epsilon/k_B T} \right].$$

Recall the identity:

$$\ln(1 + e^{-x}) = \ln \left[2 \cosh \left(\frac{x}{2} \right) \right] - \frac{x}{2}.$$

So:

$$\ln \left[1 + e^{-\epsilon/k_B T} \right] = \ln \left[2 \cosh \left(\frac{\epsilon}{2k_B T} \right) \right] - \frac{\epsilon}{2k_B T}.$$

Therefore:

$$F_{\text{bulk}}^\sigma = -k_B T \int_0^\infty d\epsilon \rho^\sigma(\epsilon, \phi^\sigma) \left[\ln \left(2 \cosh \left(\frac{\epsilon}{2k_B T} \right) \right) - \frac{\epsilon}{2k_B T} \right].$$

The zero-temperature part (i.e., $\int \rho(\epsilon)\epsilon$) is subtracted to get the thermal free energy, leading to the regularized form:

$$F_{\text{bulk}}^\sigma = -k_B T \int_0^\infty d\epsilon \rho^\sigma(\epsilon, \phi^\sigma) \ln \left[2 \cosh \left(\frac{\epsilon}{2k_B T} \right) \right].$$

1.3 Adding localized edge state contribution

If there's a localized edge mode at energy $\epsilon_s^\sigma(\phi^\sigma)$, then its contribution must be added separately, since it's not included in $\rho(\epsilon)$. The partition function of a two-level system with energy levels $\pm\epsilon/2$ is:

$$Z = e^{-\epsilon/2k_B T} + e^{\epsilon/2k_B T} = 2 \cosh \left(\frac{\epsilon}{2k_B T} \right),$$

so, the free energy for one such state is:

$$F_{\text{edge}} = -k_B T \ln \left[2 \cosh \left(\frac{\epsilon_s^\sigma(\phi^\sigma)}{2k_B T} \right) \right].$$

1.4 Combine both contributions

Adding the localized edge contribution and the bulk integral:

$$F^\sigma(\phi^\sigma, T) = F_{\text{edge}} + F_{\text{bulk}}^\sigma = -k_B T \left\{ \ln \left[2 \cosh \left(\frac{\epsilon_s^\sigma(\phi^\sigma)}{2k_B T} \right) \right] + \int_0^\infty d\epsilon \rho^\sigma(\epsilon, \phi^\sigma) \ln \left[2 \cosh \left(\frac{\epsilon}{2k_B T} \right) \right] \right\}.$$

- The log cosh term: free energy of a single localized (bound) mode.
- The integral term: continuum of edge modes with density $\rho^\sigma(\epsilon, \phi^\sigma)$.
- The factor of 2 inside the cosh reflects the particle-hole symmetry or the degeneracy between $\pm\epsilon$ modes.

2 Josephson current

2.1 Josephson current as a conjugate variable to ϕ

To prove that the Josephson current is the conjugate variable to the superconducting phase difference, we'll start from the first principles in quantum statistical mechanics and then connect it to the Josephson effect.

2.1.1 Quantum system with phase-dependent Hamiltonian

Let the Hamiltonian of the system depend on a parameter ϕ (superconducting phase difference). At temperature T , the free energy is:

$$F(\phi, T) = -k_B T \ln Z(\phi)$$

where the partition function is:

$$Z(\phi) = \text{Tr} \left[e^{-\beta H(\phi)} \right], \quad \beta = \frac{1}{k_B T}$$

2.1.2 Derivative of free energy

Differentiate $F(\phi)$ with respect to ϕ :

$$\frac{\partial F}{\partial \phi} = -k_B T \cdot \frac{1}{Z} \cdot \frac{\partial Z}{\partial \phi}$$

But:

$$\frac{\partial Z}{\partial \phi} = \text{Tr} \left[-\beta \frac{\partial H}{\partial \phi} e^{-\beta H} \right] = -\beta \cdot \text{Tr} \left[\frac{\partial H}{\partial \phi} e^{-\beta H} \right]$$

So:

$$\frac{\partial F}{\partial \phi} = \frac{1}{Z} \cdot \text{Tr} \left[\frac{\partial H}{\partial \phi} e^{-\beta H} \right] = \left\langle \frac{\partial H}{\partial \phi} \right\rangle$$

The result is:

$$\boxed{\frac{\partial F}{\partial \phi} = \left\langle \frac{\partial H}{\partial \phi} \right\rangle}$$

This is a general thermodynamic identity.

2.2 Applying conjugate variables identity to Josephson junction

In a Josephson junction, the Hamiltonian for a weak link between two superconductors depends on the phase difference ϕ between them.

For example:

$$H(\phi) = -E_J \cos(\phi)$$

Then:

$$\frac{\partial H}{\partial \phi} = E_J \sin(\phi)$$

The Josephson current is given by:

$$I = -\frac{2e}{\hbar} \left\langle \frac{\partial H}{\partial \phi} \right\rangle$$

So, combining with the thermodynamic identity:

$$I = -\frac{2e}{\hbar} \frac{\partial F}{\partial \phi}$$

Or more commonly written as:

$$I = \frac{2e}{\hbar} \frac{\partial F}{\partial \phi}$$

2.2.1 Physical meaning:

- In quantum mechanics, if a parameter λ is a coordinate, then $\partial H / \partial \lambda$ is its conjugate force (like momentum to position).
- Here, the phase ϕ is the coordinate, and current I is its conjugate observable.
- The current measures how the energy changes with phase: this is analogous to how force is the derivative of potential energy with position.

Therefore:

$$I = \frac{2e}{\hbar} \frac{\partial F}{\partial \phi} \quad (\text{current is conjugate to phase})$$